

Nose

External Nose: Nostrils (elliptical orifices), ala nasi (lateral rounded part).

Nasal Cavity: Extends from nostrils to **choanae** (posterior nasal apertures), **opens into nasopharynx**.

Regions of the Nasal Cavity

Involved in **air conditioning** (warming/humidifying), **olfaction**, and **defense**.

Clinical Notes (Nasal Cavity)

Epistaxis (nosebleed) – often from Little's (**Kiesselbach's**) area.

Rhinorrhea, Rhinitis.

Paranasal Sinuses

Air cavities: **Frontal, Ethmoidal, Sphenoidal, Maxillary.**

Functions: **warm/humidify air, resonance, lighten skull, absorb shock, facial growth.**

★ **Sinusitis – inflammation of sinus lining.**

Pharynx

★ Muscular tube (~**12-13 cm**), behind nasal/oral/laryngeal cavities. Continues **with esophagus at C6 level.**

Divided into:

1. **Nasopharynx** – **behind nasal cavity; auditory tube opens here.**
2. **Oropharynx** – **behind oral cavity.**
3. **Laryngopharynx** – **from hyoid bone to cricoid cartilage (continuous with esophagus).**

Larynx

- ★ **Phonation box** and **air passage.**
- ★ Composed of **cartilages, muscles, ligaments.**
- ★ Contains **vocal cords** (2 pairs of vocal folds).

Trachea

- ★ **10 cm long; begins at larynx and ends at carina (bifurcation)** into right and left primary bronchi.
- ★ Supported by **C-shaped hyaline cartilage rings.**

Bronchi & Bronchial Tree

- ★ Right and left **principal bronchi** arise at **sternal angle.**
- ★ Divide into **secondary and tertiary (segmental) bronchi.**

Lungs

1. Covered by **pleura**, located in **thoracic cavity**.
2. **Right lung**: 3 lobes (upper, middle, lower), 2 fissures (oblique + horizontal).
3. **Left lung**: 2 lobes, 1 oblique fissure, has **cardiac notch**.

Lung Surfaces:

- ★ **Costal** – smooth, convex (next to ribs).
- ★ **Mediastinal** – concave, contains **hilum** (entry/exit for bronchi, vessels, nerves).
- ★ **Diaphragmatic** – concave, base resting on diaphragm.

Pleura

- ★ **Serous sac** with two layers:
- ★ **Parietal pleura** (outer)
- ★ **Visceral pleura** (inner)
- ★ **Pleural cavity** lies between them.

Pleuritis: inflammation of pleura

Pneumothorax: air within pleura

Haemothorax: blood within pleura

Diaphragm

- ★ **Musculotendinous** structure with:
- ★ **Muscular part** (fixed)
- ★ **Central tendon** (movable, no bony attachment)

Diaphragmatic Openings:

- ★ Structure Level Passes Through
- ★ **IVC** (Inferior vena cava) -----> **T8 Caval opening**
- ★ **Esophagus** -----> **T10 Esophageal hiatus**
- ★ **Aorta** -----> **T12 Aortic hiatus**

Intercostal Muscles

1. **External**
2. **Internal**
3. **Innermost** (includes **intercostalis, subcostalis, transversus thoracis**)